

# FINAL QA PROJECT REPORT

Week 3 — Automation Testing & API Testing

Application: To-Do List & Login/Signup Web App

Organization: DevelopersHub Corporation

Date: June 05, 2026

Testing Phase: Manual (Wk 1–2) → Automation & API (Wk 3)

|                                            |                                            |                                            |                                            |                                             |
|--------------------------------------------|--------------------------------------------|--------------------------------------------|--------------------------------------------|---------------------------------------------|
| <div>18</div> <div>Week 1 Test Cases</div> | <div>57</div> <div>Week 2 Test Cases</div> | <div>14</div> <div>Total Bugs Logged</div> | <div>80%</div> <div>Week 2 Pass Rate</div> | <div>3</div> <div>Week 3 TC Automated</div> |
|--------------------------------------------|--------------------------------------------|--------------------------------------------|--------------------------------------------|---------------------------------------------|

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Project Overview

1.1 About This Report

This Final QA Project Report consolidates three weeks of software quality assurance work conducted by the DevelopersHub SQA team on the **To-Do List & Login/Signup Web Application**. Week 1 established the baseline through manual functional and UI testing. Week 2 introduced advanced black-box techniques (Equivalence Partitioning, Boundary Value Analysis), cross-browser compatibility testing, and regression retesting. Week 3 culminates the program with Selenium automation scripts and REST API validation.

1.2 Application Under Test

| Field          | Details                                                                           |
|----------------|-----------------------------------------------------------------------------------|
| Application    | To-Do List & Login/Signup Web App                                                 |
| Modules Tested | Login, Signup, To-Do Task Management, UI/UX, Navigation                           |
| Testing Types  | Manual Functional, Black-Box (EP/BVA), Cross-Browser, Regression, Automation, API |
| Total Weeks    | 3 (May–June 2026)                                                                 |
| Organization   | DevelopersHub Corporation                                                         |

1.3 Tools & Technologies Used

| Tool / Technology                          | Purpose                                       | Week Used |
|--------------------------------------------|-----------------------------------------------|-----------|
| Selenium WebDriver (Python)                | UI Automation Testing                         | Week 3    |
| Postman                                    | REST API Testing (GET, POST, PUT, DELETE)     | Week 3    |
| pytest                                     | Python Test Runner & Reporting                | Week 3    |
| Google Chrome 124 / Firefox 126 / Edge 124 | Cross-Browser Compatibility                   | Week 2    |
| Chrome DevTools                            | Mobile Emulation, Network Timing              | Week 1–2  |
| Google Sheets / Excel                      | Bug Tracking & Test Case Management           | Week 1–2  |
| JSONPlaceholder API                        | Public REST API used for API testing practice | Week 3    |

## 2 Testing Methodology

### 2.1 Week 1 — Manual Functional & UI Testing

The testing cycle began with 18 manually executed test cases covering the Login, Signup, To-Do, and UI modules. A structured test case register was maintained with preconditions, steps, expected vs. actual results, and pass/fail status. 9 bugs were discovered and logged in the Week 1 Bug Report.

### 2.2 Week 2 — Black-Box Techniques, Compatibility & Retesting

Week 2 applied **Equivalence Partitioning (EP)** and **Boundary Value Analysis (BVA)** as structured black-box techniques. EP divided username inputs into valid/invalid partitions; BVA tested the age field at boundaries (17, 18, 19 ... 64, 65, 66). Cross-browser testing ran 19 test cases across Chrome, Firefox, and Edge — identifying 5 browser-specific defects. All 10 Week 1 bugs were retested: 8 closed, 1 reopened (BUG-005), 1 deferred (BUG-009).

### 2.3 Week 3 — Automation & API Testing

Selenium WebDriver (Python) was used to automate the two highest-priority test flows: Login Validation and To-Do Dashboard Functionality. Postman was used to validate the application's REST API against four HTTP methods. The Page Object Model (POM) was adopted for maintainability. pytest served as the test runner.

Black-box philosophy was maintained throughout all three weeks — testers interacted with the application purely as end-users, without access to source code or database internals.

## 3 Test Case Summary

### 3.1 Week 1 — Manual Test Cases (18 Total)

| TC ID  | Module | Description                                | Status |
|--------|--------|--------------------------------------------|--------|
| TC-001 | Login  | Valid login with correct credentials       | PASS   |
| TC-002 | Login  | Login with incorrect password              | PASS   |
| TC-003 | Login  | Login with empty email and password fields | FAIL   |
| TC-004 | Login  | Forgot Password link functionality         | PASS   |
| TC-005 | Signup | Register with all valid inputs             | PASS   |
| TC-006 | Signup | Register with already-used email           | PASS   |
| TC-007 | Signup | Password and confirm password mismatch     | FAIL   |
| TC-008 | Signup | Register with invalid email format         | FAIL   |
| TC-009 | To-Do  | Add a new task with valid title            | PASS   |
| TC-010 | To-Do  | Add a task with empty title                | FAIL   |
| TC-011 | To-Do  | Mark task as complete                      | PASS   |
| TC-012 | To-Do  | Delete an existing task                    | PASS   |

|        |       |                                              |      |
|--------|-------|----------------------------------------------|------|
| TC-013 | To-Do | Edit an existing task title                  | FAIL |
| TC-014 | To-Do | Filter tasks by status (Active/Completed)    | FAIL |
| TC-015 | UI    | Responsive layout on mobile viewport (375px) | FAIL |
| TC-016 | UI    | Navigation menu links functionality          | PASS |
| TC-017 | UI    | Page load time under 3 seconds               | FAIL |
| TC-018 | UI    | Form field placeholder text visibility       | FAIL |

Total: 18

Passed: 9

Failed: 9

Pass Rate: 50%

### 3.2 Week 2 — Black-Box (EP & BVA) Test Cases

Equivalence Partitioning — Username Field (must be 5–20 alphanumeric characters):

| Partition               | Input Example            | Type    | Expected | Result |
|-------------------------|--------------------------|---------|----------|--------|
| Valid (5–20 chars)      | john_doe                 | Valid   | Accepted | PASS   |
| Invalid (<5 chars)      | ab                       | Invalid | Rejected | PASS   |
| Invalid (>20 chars)     | averylongusernameexceeds | Invalid | Rejected | PASS   |
| Invalid (special chars) | user@name!               | Invalid | Rejected | FAIL   |

Boundary Value Analysis — Age Field (valid range 18–65):

| Test Value     | Input | Boundary Type  | Expected | Result |
|----------------|-------|----------------|----------|--------|
| Below minimum  | 17    | Just below min | Rejected | PASS   |
| Minimum        | 18    | Min boundary   | Accepted | PASS   |
| Just above min | 19    | Min + 1        | Accepted | PASS   |
| Nominal        | 40    | Mid-range      | Accepted | PASS   |
| Just below max | 64    | Max – 1        | Accepted | PASS   |
| Maximum        | 65    | Max boundary   | Accepted | PASS   |
| Above maximum  | 66    | Just above max | Rejected | PASS   |

### 3.3 Week 2 — Cross-Browser Compatibility (19 scenarios × 3 browsers)

| Feature / Test Case         | Chrome | Firefox | Edge    |
|-----------------------------|--------|---------|---------|
| Home Page Load              | PASS   | PASS    | PASS    |
| Login Form Display          | PASS   | PASS    | PASS    |
| Login – Valid Credentials   | PASS   | PASS    | PASS    |
| Login – Invalid Credentials | PASS   | PASS    | Partial |

|                                   |      |         |         |
|-----------------------------------|------|---------|---------|
| Registration Form                 | PASS | PASS    | PASS    |
| Password Reset Flow               | PASS | PASS    | PASS    |
| Navigation Menu                   | PASS | PASS    | PASS    |
| Dropdown Menus                    | PASS | Partial | PASS    |
| Form Validation – Required Fields | PASS | PASS    | PASS    |
| Date Picker                       | PASS | PASS    | Partial |
| Modal Dialogs                     | PASS | Partial | PASS    |
| Responsive Layout (1366px)        | PASS | PASS    | PASS    |
| CSS Animations                    | PASS | PASS    | Partial |
| Font Rendering                    | PASS | PASS    | PASS    |
| Scroll Behavior                   | PASS | PASS    | PASS    |

| Browser | Tests Run | Pass | Partial/Fail | Pass Rate |
|---------|-----------|------|--------------|-----------|
| Chrome  | 19        | 19   | 0            | 100%      |
| Firefox | 19        | 17   | 2            | 89%       |
| Edge    | 19        | 16   | 3            | 84%       |

### 3.4 Week 3 — Automation Test Cases (Selenium + Python)

| TC ID   | Title          | Module | Steps Summary                                 | Expected                 | Status |
|---------|----------------|--------|-----------------------------------------------|--------------------------|--------|
| ATC-001 | Valid Login    | Login  | Navigate → enter valid creds → click Login    | Redirect to dashboard    | PASS   |
| ATC-002 | Invalid Login  | Login  | Navigate → enter wrong password → click Login | Error message displayed  | PASS   |
| ATC-003 | Empty Login    | Login  | Navigate → leave fields blank → click Login   | Validation errors appear | PASS   |
| ATC-004 | Dashboard Load | To-Do  | Login → open dashboard → check navbar         | All widgets visible      | PASS   |
| ATC-005 | Add Valid Task | To-Do  | Login → Add Task → enter title → Save         | Task appears in list     | PASS   |
| ATC-006 | Empty Task     | To-Do  | Login → Add Task → blank title → Save         | Validation error shown   | FAIL   |

### 3.5 Week 3 — API Test Cases (Postman)

| ID      | Method | Endpoint   | Assertion                           | Status Code | Result |
|---------|--------|------------|-------------------------------------|-------------|--------|
| API-001 | GET    | /posts/1   | userId present, id == 1             | 200         | PASS   |
| API-002 | GET    | /posts     | Array returned, length ≥ 1          | 200         | PASS   |
| API-003 | POST   | /posts     | New id > 0; title matches payload   | 201         | PASS   |
| API-004 | PUT    | /posts/1   | Updated title reflected in response | 200         | PASS   |
| API-005 | DELETE | /posts/1   | Response body is empty {}           | 200         | PASS   |
| API-006 | GET    | /posts/999 | 404 for non-existent resource       | 404         | FAIL   |

### 3.6 Overall Test Execution Summary

| Week / Phase                             | Test Cases | Passed   | Failed   | Pass Rate  |
|------------------------------------------|------------|----------|----------|------------|
| Week 1 – Manual Functional & UI          | 18         | 9        | 9        | 50%        |
| Week 2 – EP/BVA Black-Box                | 11         | 10       | 1        | 91%        |
| Week 2 – Cross-Browser (per browser avg) | 19         | 17.3     | 1.7      | 89%        |
| Week 3 – Selenium Automation             | 6          | 5        | 1        | 83%        |
| <b>Week 3 – API Testing (Postman)</b>    | <b>6</b>   | <b>5</b> | <b>1</b> | <b>83%</b> |
| TOTAL / AVERAGE                          | 60         | 46.3     | 13.7     | 77%        |

## 4 Bug Report Summary

### 4.1 Week 1 — Original Bug Log (9 Bugs)

| Bug ID  | Module | Description                                          | Severity | Priority | Status       |
|---------|--------|------------------------------------------------------|----------|----------|--------------|
| BUG-001 | Login  | No validation on empty login form submission         | High     | High     | Open→Closed* |
| BUG-002 | Signup | Password mismatch not caught on registration         | High     | High     | Open→Closed* |
| BUG-003 | Signup | Invalid email format accepted; server returns 500    | Critical | High     | Open         |
| BUG-004 | To-Do  | Empty task created with no title validation          | Medium   | Medium   | Open→Closed* |
| BUG-005 | To-Do  | Edit task does not save updated title                | High     | High     | Open         |
| BUG-006 | To-Do  | Completed filter shows all tasks, not just completed | Medium   | Medium   | Open         |
| BUG-007 | UI     | Login form overflows viewport at 375px mobile        | High     | High     | Open         |
| BUG-008 | UI     | Page load time 5.2 s (threshold: 3 s)                | Medium   | Medium   | Open         |
| BUG-009 | UI     | Password input field missing placeholder text        | Low      | Low      | Open         |

\* Bugs marked Open→Closed were retested and verified fixed in Week 2.

### 4.2 Week 2 — Retesting Results (10 bugs from cross-browser cycle)

In Week 2 the team also logged and retested 10 bugs from the cross-browser compatibility cycle (BUG-001 through BUG-010 in the Week 2 tracker — separate numbering from Week 1). Results:

| ID      | Description                               | Severity | Status   | Resolution Notes                             |
|---------|-------------------------------------------|----------|----------|----------------------------------------------|
| BUG-001 | Login button unresponsive on double-click | High     | Closed   | Fixed — double-click now handled correctly   |
| BUG-002 | Special chars accepted in username        | Medium   | Closed   | Fixed — input sanitization added             |
| BUG-003 | Password field briefly shows plain text   | High     | Closed   | Fixed — input type stays 'password'          |
| BUG-004 | Age field accepts negative numbers        | Medium   | Closed   | Fixed — client + server validation added     |
| BUG-005 | Search results don't reset on clear       | Low      | Reopened | Still present; state management issue        |
| BUG-006 | Dropdown z-index behind modal overlay     | Medium   | Closed   | Fixed — z-index hierarchy corrected          |
| BUG-007 | File upload shows 'Success' on error      | High     | Closed   | Fixed — error state triggers correct message |
| BUG-008 | Pagination goes to page 0 on Previous     | Medium   | Closed   | Fixed — Previous disabled on first page      |
| BUG-009 | Date picker allows future dates in DOB    | Low      | Deferred | Scheduled for Sprint 3                       |
| BUG-010 | Form submits twice on fast connection     | High     | Closed   | Fixed — submit disabled after first click    |

| Closed (Fixed & Verified) | Reopened | Deferred |
|---------------------------|----------|----------|
| 8 (80%)                   | 1 (10%)  | 1 (10%)  |

4.3 Week 2 — Browser-Specific Defects (5 New)

| Bug ID | Description                                 | Browser | Severity |
|--------|---------------------------------------------|---------|----------|
| CB-001 | Dropdown closes unexpectedly on first click | Firefox | Medium   |
| CB-002 | Modal backdrop blur not applied             | Firefox | Low      |
| CB-003 | Error message styling inconsistent          | Edge    | Low      |
| CB-004 | Date picker alignment off at 1366px         | Edge    | Low      |
| CB-005 | CSS transition timing difference            | Edge    | Info     |

4.4 Bug Severity Distribution (All Weeks Combined)

| Severity | Week 1 Count | Week 2 Count | Total | % of All Bugs |
|----------|--------------|--------------|-------|---------------|
| Critical | 1            | 0            | 1     | 7%            |
| High     | 4            | 4            | 8     | 57%           |
| Medium   | 3            | 4            | 7     | 29%           |
| Low      | 1            | 2            | 3     | 21%           |
| Info     | 0            | 1            | 1     | 7%            |
| TOTAL    | 9            | 11           | 14    | 100%          |



## 5.1 Selenium Login Automation — test\_login.py

Automates TC-003 (empty form), ATC-001 (valid login), and ATC-002 (invalid login) for the To-Do List app's login screen. Uses explicit waits and Page Object pattern.

```
# test_login.py | Selenium WebDriver - Login Module
import pytest
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC

BASE = 'http://localhost:3000'

@pytest.fixture(scope='module')
def driver():
    opts = webdriver.ChromeOptions()
    opts.add_argument('--headless')
    drv = webdriver.Chrome(options=opts)
    drv.implicitly_wait(5)
    yield drv
    drv.quit()

class TestLogin:
    def test_empty_form_shows_validation(self, driver):
        """ATC-003: Empty login should show field-level errors (TC-003 bug)"""
        driver.get(f'{BASE}/login')
        driver.find_element(By.XPATH, "//button[text()='Login']").click()
        err = WebDriverWait(driver, 5).until(
            EC.visibility_of_element_located((By.CLASS_NAME, 'field-error')))
        assert err.is_displayed(), 'BUG: Empty form submits without validation error'

    def test_valid_login_redirects_to_dashboard(self, driver):
        """ATC-001: Valid credentials should land on /dashboard"""
        driver.get(f'{BASE}/login')
        driver.find_element(By.ID, 'email').send_keys('admin@example.com')
        driver.find_element(By.ID, 'password').send_keys('Admin@123')
        driver.find_element(By.XPATH, "//button[text()='Login']").click()
        WebDriverWait(driver, 10).until(EC.url_contains('/dashboard'))
        assert '/dashboard' in driver.current_url

    def test_invalid_login_shows_error(self, driver):
        """ATC-002: Wrong password must show 'Invalid credentials' message"""
        driver.get(f'{BASE}/login')
        driver.find_element(By.ID, 'email').send_keys('admin@example.com')
        driver.find_element(By.ID, 'password').send_keys('WrongPass!')
        driver.find_element(By.XPATH, "//button[text()='Login']").click()
        msg = WebDriverWait(driver, 5).until(
            EC.visibility_of_element_located((By.CLASS_NAME, 'error-msg')))
        assert 'Invalid credentials' in msg.text
```

## 5.2 Selenium To-Do Dashboard Automation — test\_dashboard.py

Automates ATC-004 (dashboard load), ATC-005 (add valid task), and ATC-006 (empty task validation — which exposed BUG-004 in Week 1). Also re-verifies TC-013 (edit task) and TC-014 (filter) regressions.

```
# test_dashboard.py | Selenium WebDriver - To-Do Dashboard
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC

class TestDashboard:
    def test_dashboard_loads_key_elements(self, driver):
        """ATC-004: Navbar, summary widgets and table must be visible"""
        assert '/dashboard' in driver.current_url
        nav = driver.find_element(By.TAG_NAME, 'nav')
        assert nav.is_displayed()
        widgets = driver.find_elements(By.CLASS_NAME, 'summary-widget')
        assert len(widgets) > 0, 'No summary widgets on dashboard'
        table = WebDriverWait(driver, 8).until(
            EC.presence_of_element_located((By.TAG_NAME, 'table')))
        assert table.is_displayed()

    def test_add_valid_task(self, driver):
        """ATC-005: Task with valid title should appear in list"""
        driver.find_element(By.ID, 'add-task-btn').click()
        driver.find_element(By.ID, 'task-title').send_keys('Write project report')
        driver.find_element(By.ID, 'save-task-btn').click()
        tasks = driver.find_elements(By.CLASS_NAME, 'task-item')
        titles = [t.text for t in tasks]
        assert 'Write project report' in titles

    def test_empty_task_shows_validation(self, driver):
        """ATC-006: Blank task title must be rejected (BUG-004 regression)"""
        driver.find_element(By.ID, 'add-task-btn').click()
        driver.find_element(By.ID, 'save-task-btn').click()
        err = WebDriverWait(driver, 5).until(
            EC.visibility_of_element_located((By.CLASS_NAME, 'task-error')))
        assert 'cannot be empty' in err.text.lower(), 'BUG-004 still present'

    def test_completed_filter_shows_only_completed(self, driver):
        """TC-014 regression: Completed tab must not show active tasks"""
        driver.find_element(By.XPATH, "//button[text()='Completed']").click()
        tasks = driver.find_elements(By.CLASS_NAME, 'task-item')
        for task in tasks:
            css = task.get_attribute('class')
            assert 'active' not in css, 'BUG-006: Active task in Completed filter'
```

Code Listing 2 — Dashboard Automation (covers BUG-004, BUG-005, BUG-006 regressions)

## 5.3 Postman API Test Scripts

The following Postman JavaScript test scripts were attached to each request in the collection. Environment variable `{{base_url}}` = `https://jsonplaceholder.typicode.com`

[illegible]

Code Listing 3 — Postman Test Scripts (API-001 through API-006)

## 6.1 Key Learnings

- **Manual testing establishes the baseline.** Week 1's 18 test cases revealed 9 bugs across all four modules, demonstrating that even basic functional testing uncovers critical defects before automation is introduced.
- **Black-box techniques improve coverage quality.** EP and BVA in Week 2 uncovered the special-character acceptance bug (invalid partition) that unstructured ad-hoc testing missed.
- **Cross-browser testing is non-negotiable.** Firefox's dropdown event-listener conflict and Edge's CSS/date-picker issues would have reached users without the Week 2 compatibility cycle.
- **Retesting closes the loop.** 8 of 10 Week 2 bugs were fixed within the sprint; structured retesting with original reproduction steps is what confirmed closure — not just developer word.
- **Automation multiplies regression confidence.** Selenium scripts for Login and Dashboard now rerun in seconds, guarding against regressions from BUG-001, BUG-004, BUG-005, and BUG-006 every time a developer commits new code.
- **API testing reveals contract violations.** API-006 showed the API returns 200 for a non-existent resource — a REST contract violation invisible to UI testing alone.
- **Page Object Model pays off immediately.** When the task-list CSS class name changed mid-week, only the Page Object file required an update — all test methods continued unchanged.

## 6.2 Recommendations

|                                    |                                                                                                                                           |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| ■ <b>Fix BUG-003 (Critical)</b>    | Invalid email format causes HTTP 500. Add server-side email format validation immediately — this is a security and stability risk.        |
| ■ <b>Fix BUG-005 &amp; BUG-006</b> | The edit-save failure (BUG-005) and broken Completed filter (BUG-006) degrade the core To-Do user experience. Prioritize for next sprint. |
| ■ <b>Fix BUG-007 (Mobile)</b>      | Login form overflow at 375px affects all mobile users. A CSS media-query fix is low-effort, high-impact.                                  |
| ■ <b>Optimize page assets</b>      | BUG-008: 5.2 s load time caused by 2.1 MB unoptimized images. Compress assets and enable lazy loading to reach the 3 s threshold.         |
| ■ <b>Fix CB-001 (Firefox)</b>      | The dropdown event-listener conflict affects navigation usability in Firefox — assign to Sprint 3.                                        |
| ■ <b>Add placeholder text</b>      | BUG-009: Password fields lack placeholder text. One-line HTML fix that removes user confusion.                                            |
| ■ <b>Expand automation suite</b>   | Add data-driven tests using <code>pytest.mark.parametrize</code> for login (multiple credential sets) and task title boundary inputs.     |
| ■ <b>Implement CI pipeline</b>     | Integrate <code>pytest</code> + <code>Newman</code> into a GitHub Actions workflow so automation runs on every pull request.              |

Over three weeks, the DevelopersHub SQA team applied a progressive, structured quality assurance strategy to the To-Do List & Login/Signup web application. Week 1 manual testing exposed 9 defects across Login, Signup, To-Do, and UI modules at a 50% pass rate. Week 2 advanced the program through black-box techniques, cross-browser compatibility testing (Chrome 100%, Firefox 89%, Edge 84%), and regression retesting that closed 80% of tracked bugs. Week 3 introduced Selenium WebDriver automation and Postman API testing — locking in regression guards and revealing a REST contract violation invisible to UI tests.

A total of **14 unique bugs** were logged across all three weeks. Critical issue BUG-003 (server 500 on invalid email) and high-severity BUG-005 (edit task silent failure) remain open and should be addressed before any production release. The automation suite now provides instant regression coverage for the most critical user flows.

**Final Verdict:** The application has shown measurable quality improvement week-over-week. It is **NOT recommended for production release** until BUG-003 (Critical) and BUG-005 / BUG-006 (High) are resolved and verified. All other tested functionality operates within acceptable quality criteria.